

MATH 189 - Data Analysis

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1 Statistical Learning

1.1 Terminology

"Supervised Learning" means we have both predictors and responses.

"Predictors/input variables/features" are arranged in a vector.

$$X = (X_1, X_2, \dots, X_n)$$

"Output variables/responses" is a scalar where $y \in \mathbb{R}$

We think of data as coming from a model.

$$y = f(x) + \epsilon$$

$\begin{cases} f(x) \text{ is fixed but unknown function} \\ \epsilon \text{ is the random error term with mean zero, independent of } X \end{cases}$

Question: Why estimate f ?

Prediction: Given X , we want to predict y with $\hat{y} = \hat{f}(x)$

Mean square error is

$$\begin{aligned}\mathbb{E}(y - \hat{y})^2 &= \mathbb{E}(f(x) + \epsilon - \hat{f}(x))^2 \\ &= \mathbb{E}(f(x) - \hat{f}(x))^2 + 2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) + \mathbb{E}(\epsilon^2)\end{aligned}$$

Examine the middle term in detail

$$2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) = \mathbb{E}(f(x) - \hat{f}(x)) \cdot \mathbb{E}(\epsilon) = 0$$

We know that

$$\mathbb{E}(y - \hat{y})^2 = \mathbb{E}(f(x) - \hat{f}(x))^2 + \mathbb{E}(\epsilon^2)$$

Where the left term is reducible error and the right term is irreducible error.

"Inference" - Sometimes we want to understand the association between y and $X_1, \dots, X_p$

1. Which predictors are associated with response?
2. Is the association positive or negative?
3. Is the relationship approximately linear?

1.2 Accuracy

Training data

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$$

where $X_i = (X_{i1}, X_{i2}, \dots, X_{ip})$

n = # data points (e.g. $n = 30$)

p = # predictors (e.g. $p = 3$)

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

Defined on training data

$$\text{Test MSE} = \text{Ave } (y_0 - \hat{f}(x_0))^2$$

Defined on previously unseen data point (x_0, y_0)

Linear / affine model

$$f(x) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p$$

"Fit/train the model" usually means run an optimization on your computer that uses training data and outputs the parameters $\beta_0, \beta_1, \dots, \beta_p$ that make the MSE and hopefully the test MSE small.

Classification

Data set $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ where $X_i = (x_{i1}, \dots, x_{ip})$ and y_i is qualitative.

Error rate is the fraction of incorrect classifications.

$$\frac{1}{n} \sum_{i=1}^n \mathbb{I}\{y_i \neq \hat{y}_i\}$$

$$\mathbb{I}\{y_i \neq \hat{y}_i\} = \begin{cases} 1, & y_i \neq \hat{y}_i \\ 0, & y_i = \hat{y}_i \end{cases}$$

Test error rate

$$\text{Ave}(\mathbb{I}\{y_0 \neq \hat{y}_0\})$$

Where (x_0, y_0) is a previously unseen test data point.

K-nearest neighbors (KNN)

The KNN classifier is defined by letting $\eta_0 =$ set of indices $i \in \{1, \dots, n\}$ for K nearest neighbors of X_0 among the input data.

$\hat{y}_0 = j$ where j maximizes

$$\sum_{i \in \eta_0} \mathbb{I}\{y_i = j\}$$

which is the number of nearest neighbors (out of K) assigned label j .

2 Linear Regression

2.1 Simple Linear Regression

Underlying statistical model: $y = \beta_0 + \beta_1 X + \epsilon$ where y is the quantitative response and a single predictor X which is why it is called simple.

Prediction model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$

$\hat{\beta}_0, \hat{\beta}_1$ are estimates based on training data, where hat means estimated quantity.

Estimating coefficients

The residual sum of squares (RSS)

$$\begin{aligned}\text{RSS} &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2\end{aligned}$$

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize RSS.

$$\begin{aligned}\frac{\partial \text{RSS}}{\partial \hat{\beta}_0} &= \frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \\ &= 2 \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\ \implies \text{RSS is } &\begin{cases} \text{decreasing for really small } \hat{\beta}_0 \\ \text{increasing for really big } \hat{\beta}_0 \end{cases}\end{aligned}$$

RSS is optimal (in $\hat{\beta}_0$) when

$$\begin{aligned}0 &= \frac{1}{n} \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\ &= \hat{\beta}_0 + \hat{\beta}_1 \left(\frac{1}{n} \sum_{i=1}^n X_i \right) - \left(\frac{1}{n} \sum_{i=1}^n Y_i \right) \\ &= \hat{\beta}_0 + \hat{\beta}_1 \bar{x} - \bar{y} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}\end{aligned}$$

Where $\begin{cases} \bar{x} = \frac{1}{n} \sum_{i=1}^n X_i \text{ is the mean predictor} \\ \bar{y} = \frac{1}{n} \sum_{i=1}^n Y_i \text{ is the mean response} \end{cases}$

Covariance

$$\begin{aligned} \text{Cov}(X_i, Y_i)_{i=1}^n &= \frac{1}{n} \sum_{i=1}^n X_i Y_i - \bar{X} \bar{Y} \\ &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})(Y_i - \bar{y}) \end{aligned}$$

Variance

$$\begin{aligned} \text{Var}(X_i)_{i=1}^n &= \text{Cov}(X_i, X_i)_{i=1}^n \\ &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})^2 \end{aligned}$$

From this we can get our optimal $\hat{\beta}_1$

$$\begin{aligned} \hat{\beta}_1 &= \frac{\text{Cov}(X_i, Y_i)_{i=1}^n}{\text{Var}(X_i)_{i=1}^n} \\ &= \frac{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})(Y_i - \bar{y})}{\frac{1}{n} \sum_{i=1}^n (X_i - \bar{x})^2} \end{aligned}$$